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SMART INDIA HACKATHON 2025



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HACKATHON
2025**

Problem Statement ID: SIH25220

Problem Statement Title: Digital Contact Tracing and Screening
Tool for MDR Pathogens in Hospitals

Theme: MedTech / BioTech / HealthTech

PS Category: Software

Team ID: 70925

Team Name: CodeDivas

Problem Statement

Digital Contact Tracing and Screening Tool for MDR Pathogens in Hospitals

Problem:

- Increase of **MDR pathogens** cases in hospitals
- Manual **contact tracing**: time-consuming and error-prone
- No real-time tracking of patient and staff movements
- High risk of uncontrolled spread within hospital premises
- Lack of **integrated alerts** for infection control teams

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Data Collection



Hospital visit and
interaction

MDR clinical data form

This form is used to collect anonymized clinical information required for studying Multi-Drug Resistant (MDR) infections in hospital settings.
NOTE : No patient identity is recorded. Please provide accurate details based on your clinical records.
The data will be used only for research and model development to improve early MDR detection and infection control.

Name of the Hospital/Institution

Short answer text

Doctor's specialization / Department

- ☐ Internal Medicine
- ☐ Critical care/ ICU
- ☐ Surgery
- ☐ Emergency Medicine
- ☐ Microbiology
- ☐ Infectious Disease
- ☐ Pediatrics

Total Duration of Stay (Until Culture Sample Taken)

- ☐ < 3 days
- ☐ 3-7 days
- ☐ 8-14 days
- ☐ 15+ days

Invasive Procedures Performed

- ☐ None
- ☐ 1 procedure (central line / ventilator / urinary catheter)
- ☐ 2 procedures
- ☐ 3+ procedures

Data collected through
google forms

Proposed Solution

Develop a hospital IT add-on / web or mobile application.

Tracks patient movements and contacts inside the hospital.

Detects MDR exposure chains in real time.

Automatically flags high-risk patients and areas.

Generates instant alerts for infection-control teams.

Speeds up life-saving decisions using real-time dashboards and outbreak insights.

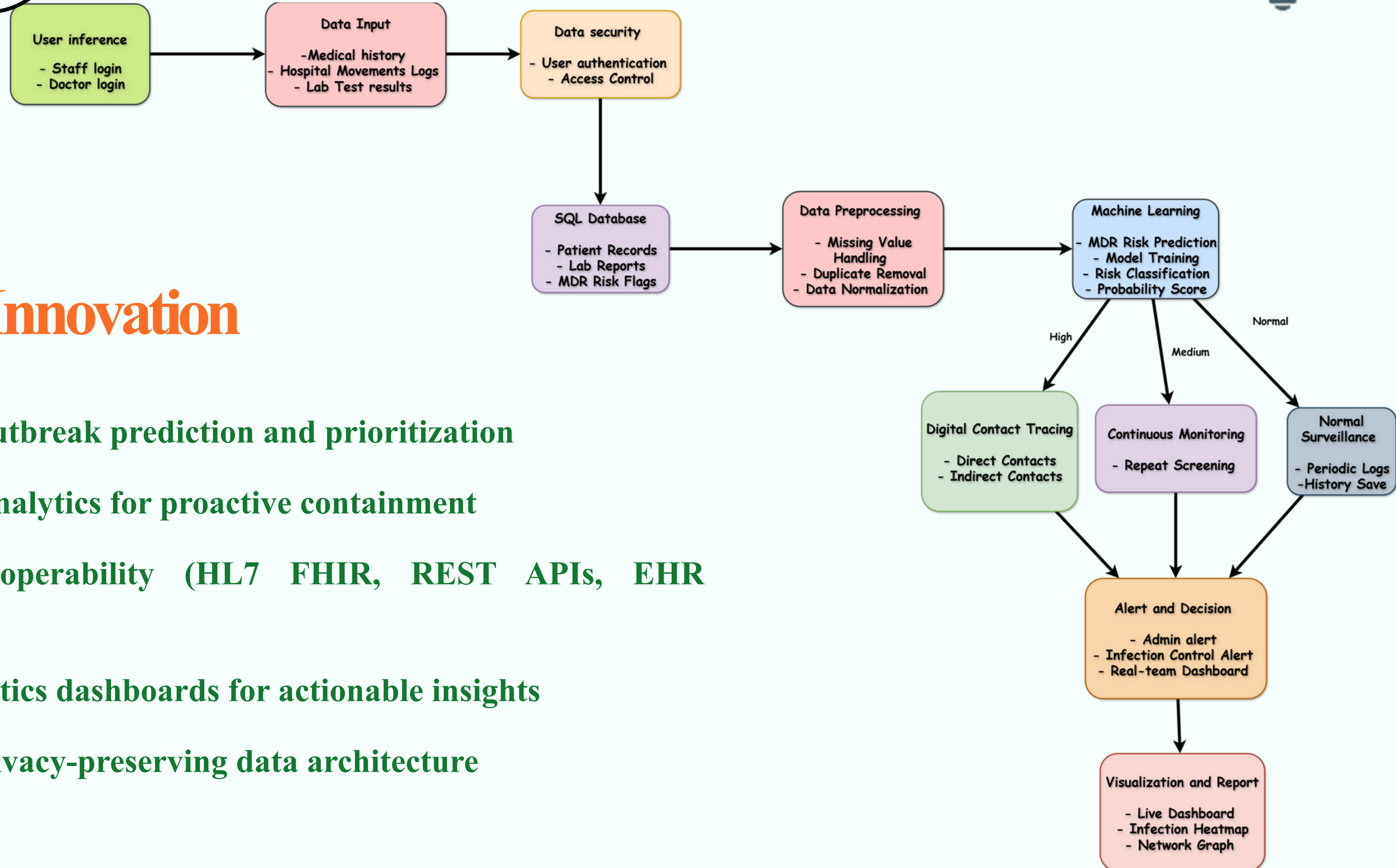
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Methodology



Innovation

- AI-driven outbreak prediction and prioritization
- Predictive analytics for proactive containment
- Open interoperability (HL7 FHIR, REST APIs, EHR integration)
- Visual analytics dashboards for actionable insights
- Scalable, privacy-preserving data architecture



Technologies Used & Architecture:

Flutter (Frontend +App Framework)

- Cross-platfrom UI for Android ,Windows,Linux,Web

Dart Language

- Single codebase for UI + backend logic

Local Database (SQLite)

- Stores patients, movements, contact and clinical data

ONNX Machine Learning Model

- MDR risk prediction executed on-device

In-App Backend Services(Dart)

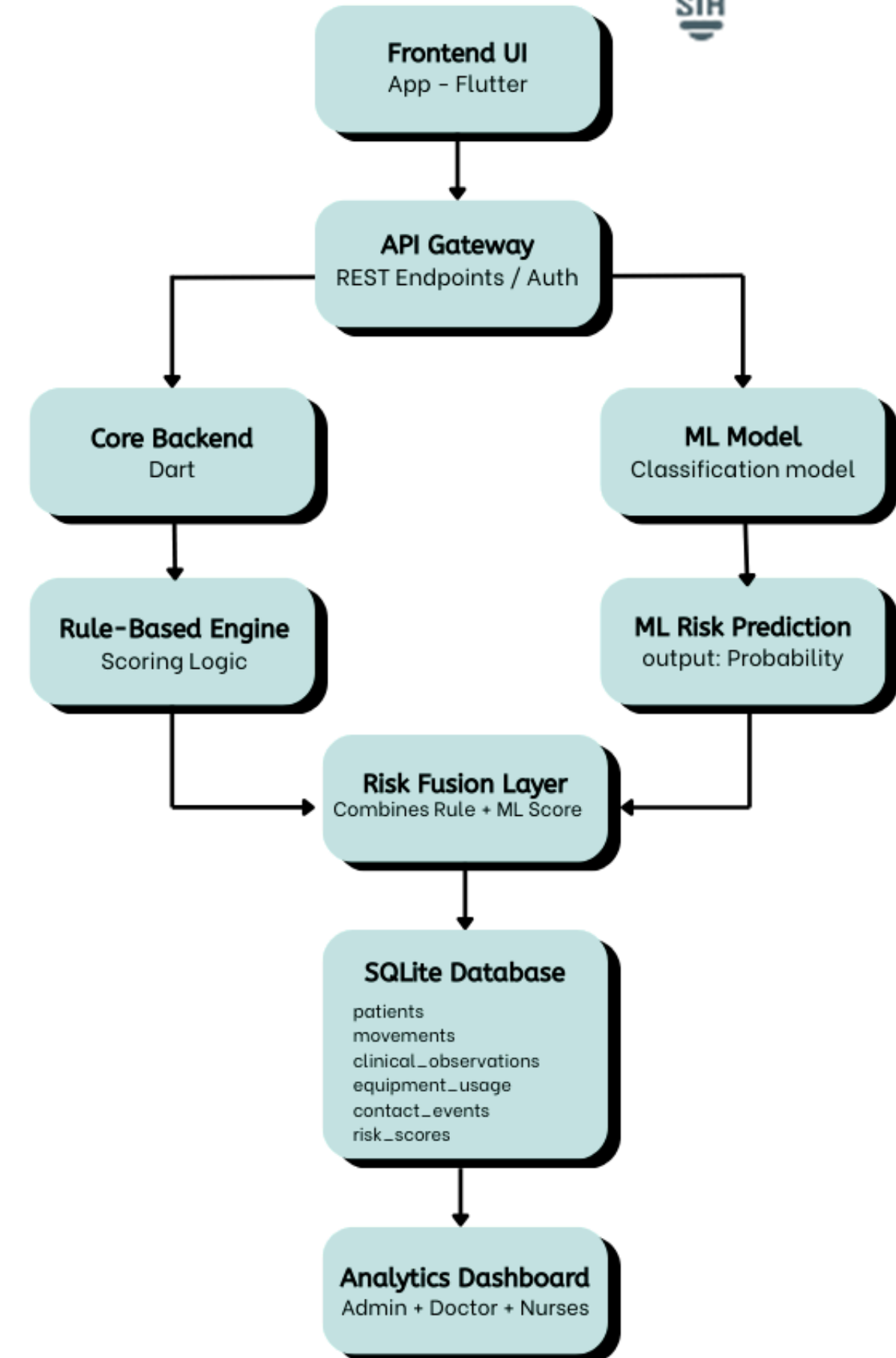
- Patient ,Movement , Contact ,Clinical

Role-Based UI

- Doctor and Nurse dashboard

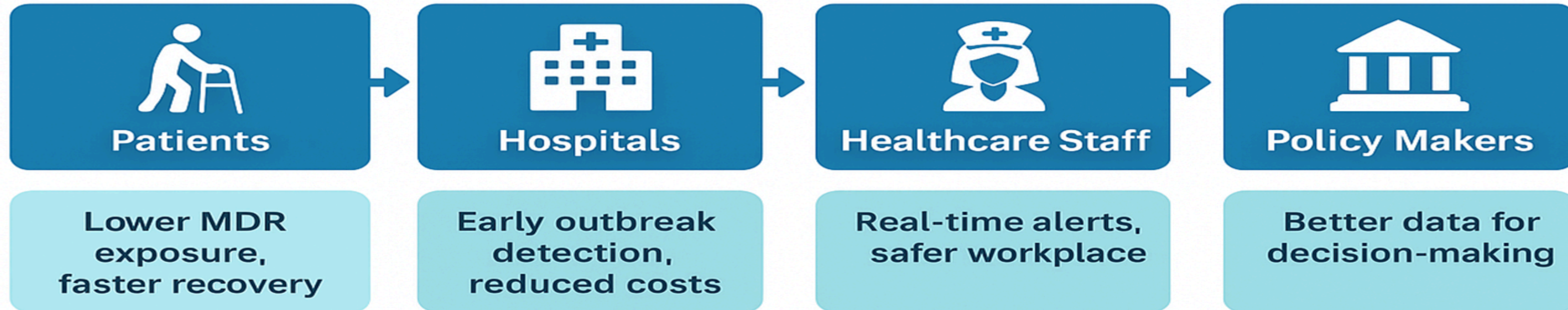
Multi-Platform Deployment

- Android mobile
- Desktop

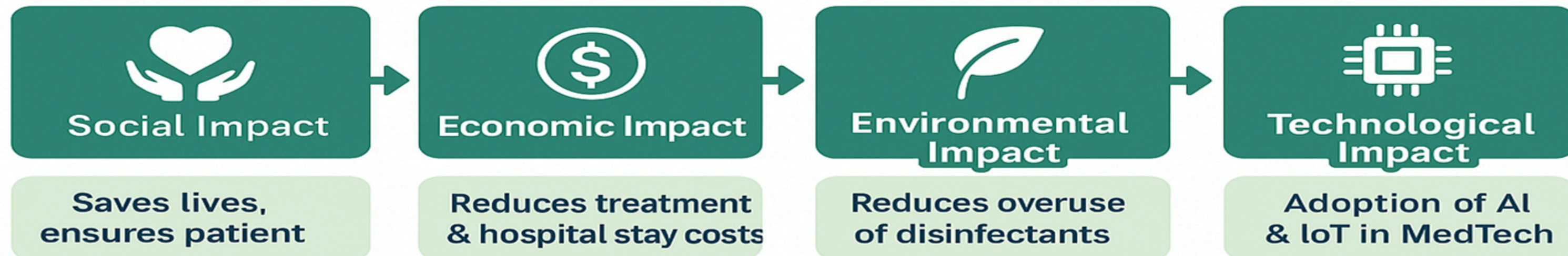


Impact and Benefits

IMPACT



BENEFITS



1. A. D. Hossain et al, "Effectiveness of contact tracing in the control of infectious diseases: a systematic review," Infect. Dis. Rep., vol. 16, no. 3, pp. 519–530, Jun. 2024.
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3. M. H. Kim et al, "Use of a real-time locating system in infection control," Infect. Control Hosp. Epidemiol., vol. 45, no. 4, pp. 1–3, Apr. 2024.
4. P. Biermann et al, "An openEHR-based infection control system to support real-time tracking of multidrug-resistant organisms," npj Digital Med., vol. 8, no. 1, pp. 1–9, 2025.
5. Z. Aklah et al, "A machine learning model for automated contact tracing in hospital settings," Comput. Biol. Med., vol. 141, pp. 105–113, 2025.
6. X. Li et al, "Uses of a real-time automatic nosocomial infection surveillance system in ICU settings," Front. Public Health, vol. 12, Article 1399067, 2024.
7. J. Rewley et al, "A passive monitoring tool using hospital administrative data for early detection of healthcare-associated infections," J. Hosp. Infect., vol. 106, no. 3, pp. 1–6, 2020

Team Member Details



Team Leader Name: M. Monikha

Branch : **BE** Stream: **CSE** Year: **III**

Team Member 1 Name: Ahana M

Branch: **BE** Stream: **AIML** Year : **III**

Team Member 2 Name: Raksha R

Branch: **BE** Stream: **CSE** Year: **III**

Team Member 3 Name: Priyanka H

Branch :**BE** Stream: **ISE** Year: **III**

Team Member 4 Name: Thanuja M

•Branch: **BE** Stream: **ECE** Year: **III**

Team Member 5 Name: Nandu D

•Branch: **BE** Stream: **CSE** Year: **III**

Team Mentor 1 Name: Prof. Priyanka S

- Category (Academic/Industry): **Academic** Years of Experience: **8 years**

Team Mentor 2 Name: Dr. Chaitra M

- Category (Academic/Industry): **Academic** Years of Experience: **16 years**

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Thank you !